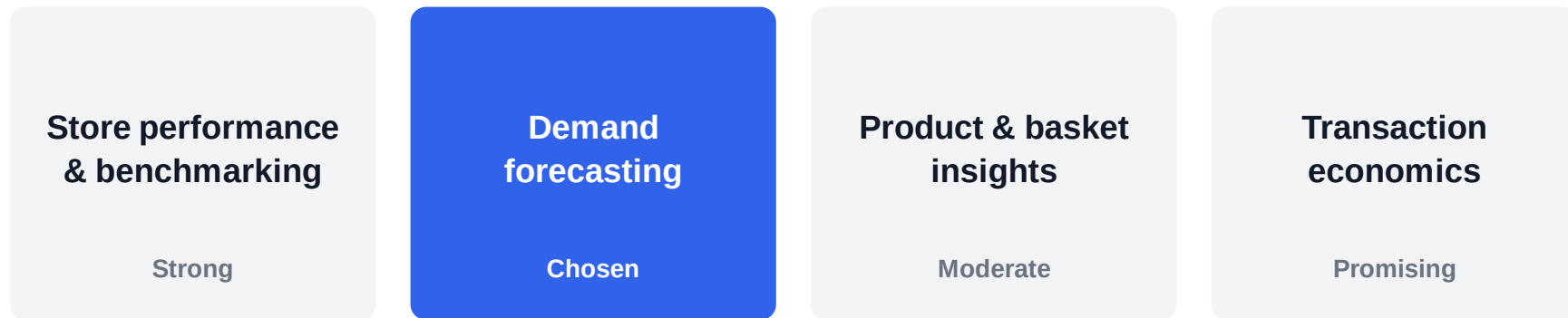


Tiller by SumUp

Can Tiller forecast next week's demand — and tell managers when to trust it?

What five years of transaction data tells us about a forecasting product

We chose demand forecasting as the focus



Forecasting combined a clear measurable signal, a plausible planning use case, and feasibility within the project.

The goal was not a model. It was evidence for a product decision.

We had enough history to test short-term forecasting

1.28M

Orders

21

Stores

5 years

2015–2020

Weekly

Demand view

We forecast weekly order volume for each store, one and two weeks ahead.

Enough history to ask whether a short-term forecast could support planning decisions.

One-week forecasts are more accurate than two-week



Forecast error (%) across the evaluation period. Lower is better.

WHAT THIS MEANS

Lower error is better.

One week is the stronger starting point for a product.

The simplest method performed best



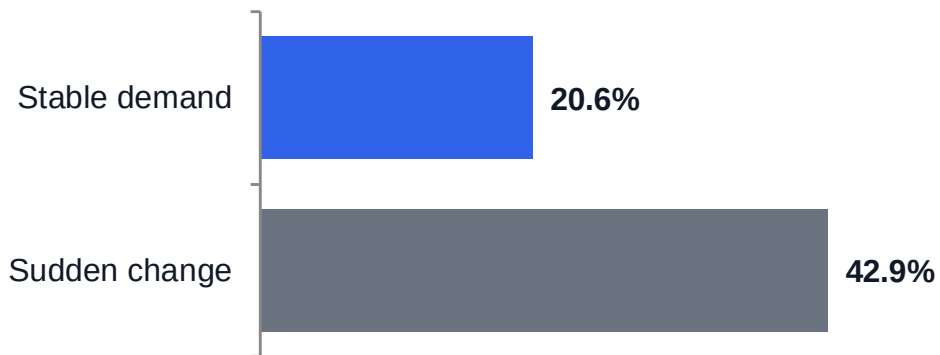
Forecast error (%) — lower is better. Axis starts at zero.

WHAT THIS MEANS

Using last week's demand beat the more complex approaches we tested.

Start simple. We do not have evidence that complexity improves accuracy for v1.

Error roughly doubles when demand changes a lot



Forecast error (%) — lower is better. Axis starts at zero.

THE DISCOVERY

20.6% → 42.9%

17 of 20 stores were less accurate when demand jumped.

A number alone can mislead. Managers need to know when to trust it less.

The signal is useful — but not the same everywhere

17 of 20 stores

improved when we added simple calendar information (e.g. day of week).

Three stores got worse. Overall error can move the other way because large stores weigh more in the total.

Forecast quality varies by store. The product should not present every forecast with equal confidence.

So what does this mean for the product?

Managers need to plan ahead, but reporting only shows what already happened. The opportunity: pair a forecast with a reliability signal, so managers know what to expect next week and how much to trust it.

1

Forecast

One-week demand is predictable enough to be useful.

2

Keep it simple

A historical baseline is enough for a first version.

3

Show reliability

Accuracy drops when demand changes. That must be visible.

Product hypothesis: better planning with forecast + reliability

Help store managers answer two questions:

1 What demand should I expect next week?

A one-week store-level forecast from recent history.

2 How much should I trust that number?

A clear signal when recent demand has changed a lot.

Not a finished design. A testable hypothesis: would this change a real planning decision?

Examples of decisions to validate: staffing, stock, weekend prep — not claimed, only to be tested.

The data supports the hypothesis. Customers decide if it's valuable.

DATA SUPPORTS

- Short-term store forecasts are feasible
- One week is stronger than two
- A simple method is enough for v1
- Reliability changes with demand
- Sudden jumps hurt accuracy a lot

PRODUCT MUST STILL VALIDATE

- What decision would the forecast change?
- Would managers use it?
- Is a reliability signal understandable?
- Which horizon matters most?
- How do we show uncertainty safely?

Next step: validate the decision — not more model complexity.